

6 Historical Counting Systems

6.4 Mayan Counting System

The Mayans were perhaps the most advanced civilizations of the ancient world. They had a sophisticated ritual system developed by a class of priests, which held time as divine and eternal. That being so, the calendar, and calculations related to it, were very important to them.

There were two numeral systems that the Mayans had, one used by priests and one used for day-to-day calculations. For the priests, the number system was governed by ritual, the days of the year were thought to be gods, so the formal symbols for the days were decorated heads, like the one below.



We're not going to study this one.

The other system they used was a base 20 positional system. The symbols they used were as follows:

Number	Vertical Form	Number	Vertical Form
0		10	
1		11	
2		12	
3		13	
4		14	
5		15	
6		16	
7		17	
8		18	
9		19	

Convert 1024 to the mayan base 20 system.

Convert 734 to the mayan base 20 system.

Convert 192 to the mayan base 20 system.

Compute $734 + 192$ in the mayan base 20 system (use the conversions above).

Compute $1028 + 734$ in the mayan base 20 system (use the conversions above).